

Notes: Carletti, Leonello, and Marquez (JFE, 2023)

Loan Guarantees, Bank Underwriting Policies and Financial Stability

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	This is a theory paper: the key objects are model primitives	4
2.2	Agents, dates, and the bank balance sheet	4
2.3	Project technology, liquidation, and shocks	5
2.4	Depositor information and run decisions	5
2.5	Guarantee types studied in the paper	6
2.6	Key endogenous thresholds	6
3	Models and empirical specifications (all equations + interpretation)	6
3.1	Project return, monitoring, and liquidation	6
3.2	Depositor information and the run game	7
3.3	Run thresholds without guarantees	7
3.4	Bank optimization without guarantees	8
3.5	First-loss loan guarantees with full bankruptcy costs	8
3.6	Bankruptcy-protected first-loss guarantees	9
3.7	Loss-sharing guarantees and scheme comparison	10
3.8	Evergreening and inefficient liquidation	10
3.9	Extensions: endogenous deposit rates and deposit insurance	11
4	Identification and instruments (what makes the estimates credible)	12
4.1	What makes the main mechanism work	12
4.2	Why bank capital matters so much	12
4.3	Why loan guarantees differ from deposit insurance	12

4.4	Why bankruptcy treatment changes the answer	12
4.5	What assumptions are doing the work	13
4.6	Relation to the literature	13
5	Figures	13
5.1	Figure 1: Bank payoff without guarantees	13
5.2	Figure 2: Depositor payoff differential when $1 - k \leq L$	14
5.3	Figure 3: Depositor payoff differential when $1 - k > L$	14
5.4	Figure 4: Bank payoff with a first-loss guarantee	14
5.5	Figure 5: Evergreening without and with a guarantee	14
6	Short summary	15
7	Conclusion	15
7.1	Core conclusion	15
7.2	Economic intuition	15
7.3	Takeaway	16

1 Big picture

- **Question.** How do **public loan guarantees** affect bank underwriting incentives and financial fragility when banks finance long-term risky loans with **demandable deposits** and can therefore be subject to runs?
- **Setting.** The paper builds a **three-date model of financial fragility** in the spirit of Goldstein–Pauzner, but adds two features that are central for banking policy:
 - banks choose costly **underwriting / monitoring effort** q ,
 - banks fund themselves with both **inside capital** k and deposits $1 - k$.

The paper then studies how different guarantee designs change run risk, effort, and inefficient loan continuation.

- **Core challenge.** The standard moral-hazard intuition says that an insurance-like guarantee should weaken incentives and encourage excessive risk taking. But for banks, loan guarantees also change the **liability side**: they affect solvency, depositor run incentives, and the bank's charter value. So the net effect is ambiguous ex ante.
- **Main theoretical contribution.** The paper embeds a bank incentive problem into a run model and shows that **asset-side guarantees** and **liability-side guarantees** work very differently. This lets the paper isolate the feedback:

loan guarantee $\rightarrow$ lower run risk / higher survival value $\rightarrow$ different underwriting incentives.

- **Why it matters.** Public guarantee schemes were used heavily during crisis episodes such as Covid-19, and they are also common in programs such as SBA lending and mortgage markets. Policy makers need to know whether such schemes merely subsidize weak lending or instead stabilize credit provision.
- **Main theoretical questions.**
 - When do loan guarantees improve monitoring?
 - How does the answer depend on **bank capital**?
 - How does it depend on whether guaranteed transfers are **lost in bankruptcy** or **bankruptcy-protected**?
 - How do **first-loss** and **loss-sharing** guarantees compare?
 - What happens to **evergreening** and to overall output?
- **Main results.**
 - In the baseline model, well-capitalized banks face only **fundamental runs**, while poorly capitalized banks are also exposed to **panic runs**.
 - A **first-loss loan guarantee with full bankruptcy costs** lowers the run threshold and **raises monitoring effort** for all banks.
 - A **bankruptcy-protected first-loss guarantee** still lowers run risk, but for very weakly capitalized banks it can **reduce monitoring effort** because it weakens depositor discipline too much.

- A **loss-sharing guarantee** gives qualitatively similar comparative statics, but for a given run threshold the **first-loss scheme is more expensive** for the government and gives **stronger incentives**.
- Loan guarantees worsen **evergreening** incentives, but when monitoring-cost curvature c is small the positive underwriting effect dominates and **total output rises**.
- If deposit rates can adjust after the guarantee is introduced, banks reduce r_2 , which **further raises monitoring effort**.
- **Deposit insurance** is different: it never improves monitoring; for poorly capitalized banks it lowers monitoring effort. If deposit insurance is already present, adding a loan guarantee still raises monitoring.
- **Economic takeaway.** Loan guarantees are **not** just deposit insurance on the asset side. Because they can increase the value of bank survival and interact with depositor runs, they may strengthen underwriting incentives—except when banks are very weak and the guarantee is insulated from bankruptcy losses.

2 Data and measurement (key objects)

2.1 This is a theory paper: the key objects are model primitives

- There is **no paper-specific empirical dataset**. The objects that matter are:
 - bank capital k ,
 - deposits $1 - k$,
 - underwriting effort q ,
 - fundamentals θ ,
 - liquidation value L ,
 - promised deposit repayment r_2 ,
 - run thresholds,
 - guarantee size and design.
- The paper has five conceptual layers:
 1. the **baseline economy without guarantees**,
 2. **first-loss guarantees** with full bankruptcy costs,
 3. **first-loss guarantees** protected from bankruptcy,
 4. **loss-sharing guarantees**,
 5. **evergreening** and extensions with endogenous deposit rates and deposit insurance.

2.2 Agents, dates, and the bank balance sheet

- Time is $t = 0, 1, 2$.

- On the **asset side**, each firm has unit demand for one risky long-term bank loan.
- On the **liability side**, the bank has:
 - **inside capital** k ,
 - **demandable deposits** of size $1 - k$.
- Depositors are atomistic, risk-neutral investors who can either:
 - withdraw at $t = 1$ for par,
 - or wait until $t = 2$ and receive $r_2 > 1$, provided the bank remains solvent.
- The bank maximizes expected profits accruing to inside capital. This makes bank capital central both for bank incentives and for fragility.

2.3 Project technology, liquidation, and shocks

- The loan finances a long-term project that yields $R\theta$ with probability q and 0 with probability $1 - q$. The bank chooses q through costly underwriting.
- Fundamentals θ summarize the macro state. They are drawn:
 - from $U[0, 1]$ with probability α ,
 - from $U[1, 2]$ with probability $1 - \alpha$.

An increase in α is interpreted as a **negative macro shock**.

- The project can be liquidated early at $t = 1$. The liquidation value is:
 - $L < 1$ for low enough fundamentals,
 - 1 for high enough fundamentals.
- Bank monitoring effort is privately costly, with cost $\frac{cq^2}{2}$.

2.4 Depositor information and run decisions

- At $t = 1$, each depositor observes a noisy private signal

$$s_i = \theta + \varepsilon_i,$$

where ε_i is uniformly distributed on $[-\varepsilon, +\varepsilon]$.

- Depositors use the signal to infer both:
 - the project's payoff prospects,
 - the likely actions of other depositors.
- This creates two possible kinds of runs:
 - **fundamental runs**, driven purely by bad fundamentals,
 - **panic runs**, driven by strategic complementarities in withdrawals.

2.5 Guarantee types studied in the paper

- **First-loss loan guarantee x .** The government covers loan losses up to amount Rx .
- **Two bankruptcy treatments for the first-loss guarantee.**
 - **Full bankruptcy costs:** the guaranteed transfer is dissipated if the bank is insolvent.
 - **Bankruptcy-protected:** the guaranteed transfer can still be used to repay depositors even when the bank is insolvent.
- **Loss-sharing guarantee y .** The government covers a fraction y of losses.
- **Deposit insurance δ .** Depositors are guaranteed a minimum repayment δ .

2.6 Key endogenous thresholds

- $\theta(k)$: fundamental-run threshold for well-capitalized banks.
- $\theta^*(q, k, L, r_2)$: panic-run threshold for weakly capitalized banks.
- $\theta_x^R, \theta_x^{*P}, \theta_y^R, \theta_\delta^*$: run thresholds under different guarantee designs.
- θ_L^B : bank's private liquidation threshold.
- θ_L^{SP} : social planner's liquidation threshold.

3 Models and empirical specifications (all equations + interpretation)

3.1 Project return, monitoring, and liquidation

The bank funds a long-term risky project whose date-2 payoff is

$$\tilde{P} = \begin{cases} R\theta, & \text{with probability } q, \\ 0, & \text{with probability } 1 - q. \end{cases} \quad (1)$$

Monitoring / underwriting effort $q \in [0, 1]$ is privately costly:

$$\text{cost of effort} = \frac{cq^2}{2}. \quad (2)$$

The project can be liquidated early at $t = 1$, yielding value $L < 1$ in bad states and 1 in sufficiently good states.

Interpretation.

- q is the bank's underwriting standard.
- Higher q makes repayment more likely, but it is costly.
- The possibility of early liquidation is what makes depositor withdrawals potentially destructive.

3.2 Depositor information and the run game

Each depositor observes

$$s_i = \theta + \varepsilon_i, \quad \varepsilon_i \sim U[-\varepsilon, \varepsilon]. \quad (3)$$

If a fraction n of depositors runs, the utility gain from waiting until date 2 rather than withdrawing at date 1 is

$$v(\theta, n) = \begin{cases} qr_2 - 1, & 0 \leq n \leq \hat{n}(\theta), \\ -1, & \hat{n}(\theta) < n \leq \bar{n}, \\ -\frac{L}{(1-k)n}, & \bar{n} < n \leq 1, \end{cases} \quad (4)$$

where $\hat{n}(\theta)$ solves

$$R\theta \left(1 - \hat{n}(\theta) \frac{1-k}{L}\right) - (1 - \hat{n}(\theta))(1-k)r_2 = 0, \quad (5)$$

and $\bar{n}$ solves

$$L = \bar{n}(1-k). \quad (6)$$

Interpretation.

- When the bank has enough capital, a depositor's incentive to run depends only on fundamentals.
- When capital is low, the payoff from waiting falls as more depositors withdraw, so strategic complementarities create panic-run risk.

3.3 Run thresholds without guarantees

The paper shows that depositor behavior has a unique threshold equilibrium.

If the bank is sufficiently well capitalized, $1 - k \leq L$, only fundamental runs occur and the run threshold is

$$\theta(k) = \frac{(1-k)r_2}{R}, \quad (7)$$

with $\partial\theta(k)/\partial k < 0$.

If the bank is weakly capitalized, $1 - k > L$, panic runs also occur below

$$\theta^*(q, k, L, r_2) = \theta \frac{qr_2 - \pi_1}{qr_2 - \pi_1 \left(\frac{1-k}{L}\right)}, \quad (8)$$

where

$$\pi_1 = \int_0^{\bar{n}} 1 \, dn + \int_{\bar{n}}^1 \frac{L}{(1-k)n} \, dn. \quad (9)$$

Interpretation.

- Higher capital k reduces the need to liquidate assets and therefore lowers run risk.
- Higher monitoring effort q lowers θ^* because depositors expect a higher date-2 payoff.
- This is the core feedback in the paper: better underwriting reduces fragility.

3.4 Bank optimization without guarantees

Let θ^R denote the relevant run threshold:

$$\theta^R = \begin{cases} \theta, & \text{if } 1 - k \leq L, \\ \theta^*, & \text{if } 1 - k > L. \end{cases} \quad (10)$$

The bank chooses q and r_2 to maximize expected profits

$$\max_{q, r_2} \alpha \int_0^{\theta^R} q \max \left\{ R\theta \left(1 - \frac{1-k}{L} \right), 0 \right\} d\theta + \alpha \int_{\theta^R}^1 q [R\theta - (1-k)r_2] d\theta + (1-\alpha) \int_1^2 q [R - (1-k)r_2] d\theta - \frac{cq^2}{2}, \quad (11)$$

subject to depositors' participation:

$$\alpha \int_0^{\theta^R} \min \left\{ \frac{L}{1-k}, 1 \right\} d\theta + \alpha \int_{\theta^R}^1 qr_2 d\theta + (1-\alpha) \int_1^2 qr_2 d\theta \geq 1. \quad (12)$$

The first-order condition for q is:

when $1 - k \leq L$,

$$\alpha \int_0^{\theta} R\theta \left(1 - \frac{1-k}{L} \right) d\theta + \alpha \int_{\theta}^1 [R\theta - (1-k)r_2] d\theta + (1-\alpha) \int_1^2 [R - (1-k)r_2] d\theta - cq = 0; \quad (13)$$

when $1 - k > L$,

$$\alpha \int_{\theta^*}^1 [R\theta - (1-k)r_2] d\theta + (1-\alpha) \int_1^2 [R - (1-k)r_2] d\theta - \alpha \frac{\partial \theta^*}{\partial q} q [R\theta^* - (1-k)r_2] - cq = 0. \quad (14)$$

Interpretation.

- For well-capitalized banks, better monitoring only raises project payoff directly.
- For weakly capitalized banks, better monitoring also lowers the run threshold, so it has an extra stability benefit.
- The paper also shows that when $1 - k \leq L$, the deposit rate r_2 is just high enough to satisfy depositor participation. When $1 - k > L$, the bank may set a higher r_2 because doing so can reduce the panic threshold.

3.5 First-loss loan guarantees with full bankruptcy costs

The government provides a first-loss guarantee x , with

$$x < \frac{1-k}{R}, \quad (15)$$

so that the guarantee alone does not fully remove depositor discipline.

With the guarantee, the bank's payoff becomes $R(\theta + x)$ in bad repayment states and R once $\theta \geq 1 - x$. The run threshold shifts to

$$\theta_x^R = \theta^R - x. \quad (16)$$

The bank's problem becomes

$$\begin{aligned} \max_q \alpha \int_0^{\theta_x^R} q \max \left\{ R(\theta + x) \left(1 - \frac{1-k}{L} \right), 0 \right\} d\theta + \alpha \int_{\theta_x^R}^{1-x} q [R(\theta + x) - (1-k)r_2] d\theta \\ + \alpha \int_{1-x}^1 q [R - (1-k)r_2] d\theta + (1-\alpha) \int_1^2 q [R - (1-k)r_2] d\theta - \frac{cq^2}{2}. \end{aligned} \quad (17)$$

Main result.

$$\frac{dq_x^R}{dx} > 0. \quad (18)$$

Interpretation.

- The guarantee reduces run risk directly by lowering the threshold from θ^R to θ_x^R .
- It also raises the bank's continuation value and therefore its **charter value**.
- Because the guarantee only pays off when the monitored loan succeeds, it strengthens monitoring incentives for all banks.

3.6 Bankruptcy-protected first-loss guarantees

Now suppose the guaranteed transfer is protected from bankruptcy losses and can still reach depositors when the bank is insolvent.

For well-capitalized banks, the run threshold remains

$$\theta_x^P = \theta_x = \theta - x. \quad (19)$$

For weakly capitalized banks, the relevant panic threshold satisfies

$$\theta_x^{*P} < \theta_x^* < \theta^*, \quad (20)$$

so the guarantee is even more effective at preventing runs than under full bankruptcy costs.

The effect on monitoring is:

$$\frac{dq_x^P}{dx} > 0 \quad \text{when } 1 - k \leq L, \quad (21)$$

but there exists a cutoff $\bar{k}_x^P < 1 - L$ such that

$$\frac{dq_x^{*P}}{dx} < 0 \quad \text{for } k < \bar{k}_x^P. \quad (22)$$

Interpretation.

- The guarantee now protects depositors even when bank monitoring fails.
- That reduces the sensitivity of depositor discipline to the bank's effort choice.
- For very weak banks, this discipline effect dominates, so monitoring can fall.

3.7 Loss-sharing guarantees and scheme comparison

Under a loss-sharing scheme, the government covers fraction $y \in (0, 1)$ of losses, so the per-unit loan return is

$$\max\{R, R\theta + R(1 - \theta)y\}. \quad (23)$$

For the scheme-comparison exercise, the paper focuses on the tractable case of **well-capitalized banks** with $1 - k \leq L$.

With full bankruptcy costs, the run threshold becomes

$$\theta_y^R = \frac{\theta^R - y}{1 - y}. \quad (24)$$

Qualitatively, the model delivers the same pattern as under first-loss guarantees:

- run thresholds fall,
- monitoring rises under full bankruptcy costs,
- and under bankruptcy protection monitoring can fall for the most weakly capitalized banks.

To compare the first-loss and loss-sharing schemes, the paper sets y so that both produce the same run threshold:

$$\frac{\theta - y}{1 - y} = \theta - x \implies y = \frac{x}{1 - \max\{\theta - x, 0\}} \geq x. \quad (25)$$

Interpretation.

- At the same run threshold, the **first-loss scheme** provides stronger incentives to monitor.
- But it also requires a **larger government disbursement**.
- So the paper does not find that one design dominates on every margin.

3.8 Evergreening and inefficient liquidation

To study zombie lending / inefficient continuation, the bank is allowed at date 1 to liquidate or continue the loan.

The bank's private liquidation threshold is

$$L - (1 - k)r_2 = q(R\theta - (1 - k)r_2),$$

so

$$\theta_L^B = \frac{L - (1 - q)(1 - k)r_2}{qR}. \quad (26)$$

The social planner liquidates when

$$L = q\theta R,$$

hence

$$\theta_L^{SP} = \frac{L}{qR}. \quad (27)$$

Since

$$\theta_L^B < \theta_L^{SP}, \quad (28)$$

banks liquidate too little and therefore engage in **evergreening**.

With a first-loss guarantee,

$$\theta_{Lx}^B = \theta_L^B - x. \quad (29)$$

Interpretation.

- The guarantee makes continuation privately more attractive, so it worsens evergreening holding q fixed.
- But the same guarantee also raises q , which offsets some of the inefficient continuation problem.
- The paper's net result is that for small enough c , the positive monitoring effect dominates and **total output rises**.

3.9 Extensions: endogenous deposit rates and deposit insurance

Endogenous deposit rates. If the existence of the guarantee is known when deposits are raised, banks optimally lower the promised deposit rate:

$$\frac{\partial r_2}{\partial x} < 0, \quad \frac{\partial q}{\partial x} > 0. \quad (30)$$

Interpretation. Loan guarantees make deposits safer, so the bank can promise less to depositors. That raises the private return to monitoring even more.

Deposit insurance only. With deposit insurance δ , the panic-run threshold for weakly capitalized banks is

$$\theta_\delta^*(q, k, \delta) = \frac{(1-k)r_2}{R} \frac{(qr_2 - \pi_1) + \delta(1-q)}{qr_2 - \pi_1 \left(\frac{1-k}{L}\right) + \delta \left(\frac{1-k}{L} - q\right)}. \quad (31)$$

The effect on monitoring is

$$\frac{dq_\delta}{d\delta} = 0 \quad \text{when } 1-k \leq L, \quad \frac{dq_\delta^*}{d\delta} < 0 \quad \text{when } 1-k > L. \quad (32)$$

Interpretation. Deposit insurance lowers run risk, but unlike a loan guarantee it does *not* raise the bank's payoff when monitoring succeeds. So it never improves monitoring and can reduce it.

Deposit insurance plus loan guarantees. When both are present, the weak-bank run threshold satisfies

$$\theta_{\delta x}^*(q, \delta) = \theta_\delta^* - x. \quad (33)$$

The effect of a first-loss loan guarantee remains positive:

$$\frac{dq_{\delta x}}{dx} > 0, \quad \frac{dq_{\delta x}^*}{dx} > 0. \quad (34)$$

Interpretation. The key force behind loan guarantees—raising the value of bank survival on the asset side—continues to operate even when deposits are already insured.

4 Identification and instruments (what makes the estimates credible)

4.1 What makes the main mechanism work

- The paper's key mechanism is the feedback between:
 - the bank's **underwriting choice** q ,
 - the bank's **solvency** / **charter value**,
 - and depositors' **run behavior**.
- Monitoring does not just affect expected loan payoff. It also changes the **run threshold**. That is what makes bank fragility and bank incentives jointly determined rather than separable.

4.2 Why bank capital matters so much

- Capital determines whether runs are purely fundamental or also panic-based.
- When $1 - k \leq L$, depositor payoffs do not depend on how many others run, so the model behaves like a standard solvency threshold problem.
- When $1 - k > L$, strategic complementarities appear and the bank becomes vulnerable to panic runs. This is exactly the region where guarantee design becomes subtle.

4.3 Why loan guarantees differ from deposit insurance

- **Loan guarantees** act on the **asset side**. They raise the bank's payoff in states where the loan would otherwise default, and so can increase the value of prudent monitoring.
- **Deposit insurance** acts on the **liability side**. It reassures depositors but does not improve the bank's project payoff.
- This is why the paper can generate:
 - positive incentive effects from loan guarantees,
 - but negative or zero incentive effects from deposit insurance.

4.4 Why bankruptcy treatment changes the answer

- If the guaranteed transfer is **lost in bankruptcy**, the bank benefits only when monitoring succeeds and the bank survives. That preserves discipline and raises effort.
- If the guaranteed transfer is **bankruptcy-protected**, depositors are cushioned even when monitoring fails. This weakens the disciplining role of runs.
- Therefore bankruptcy law and guarantee design are inseparable in the model.

4.5 What assumptions are doing the work

- The guarantee is deliberately bounded by $x < (1 - k)/R$, so it does not fully eliminate depositor discipline.
- The crisis shock is modeled as an increase in α , which worsens the distribution of fundamentals.
- The loan rate R is fixed in the baseline when guarantees are introduced unexpectedly; the extension with endogenous r_2 shows the main underwriting result is reinforced, not overturned.
- Bank capital is exogenous. The paper is explicit that endogenizing capital structure is left for future work.

4.6 Relation to the literature

- Relative to classic bank-run models, the paper adds a **bank effort choice** and internal capital.
- Relative to bank moral-hazard models, it adds **run risk** and endogenous depositor discipline.
- Relative to the public-guarantee literature, it shows that the welfare and incentive effects of guarantees depend crucially on whether they resemble **deposit insurance** or **loan guarantees**.
- Relative to the zombie-lending literature, it formalizes how guarantees can worsen inefficient continuation even while improving ex ante underwriting.

5 Figures

5.1 Figure 1: Bank payoff without guarantees

- The bank receives $R\theta$ when $\theta < 1$ and the full repayment R when $\theta \geq 1$.
- Read this figure as the paper's basic asset-side object: the bank is a lender holding debt, so its payoff is capped above and linear below full repayment.
- This capped payoff is why default states matter so much for both guarantees and monitoring incentives.

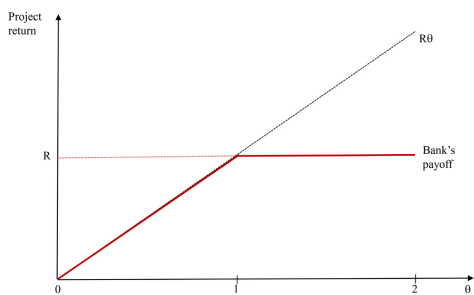


Fig. 1. Bank payoff. The figure illustrates a bank's payoff as a function of the fundamental of the economy θ . This is equal to R for any $\theta \geq 1$ and $R\theta$ for any $\theta < 1$.

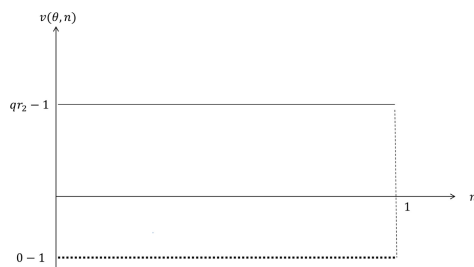


Fig. 2. Depositors' payoff differential when $(1 - k) \leq 1$. The figure shows a depositor's payoff differential between withdrawing at date 2 and at date 1 as a function of the proportion of depositors withdrawing early, n , when $1 - k \leq 1$. The payoff differential is independent of n and only depends on whether the bank is solvent at date 2. The solid line represents the utility differential when the bank is solvent at date 2, while the dotted one captures the utility differential when the bank does not have enough resources to repay depositors even if no one runs.

5.2 Figure 2: Depositor payoff differential when $1 - k \leq L$

- Here the depositor's payoff differential from waiting rather than withdrawing is **independent** of how many other depositors run.
- So there are no panic complementarities; runs are purely **fundamental**.
- The figure is visual proof of why high capital eliminates the coordination problem.

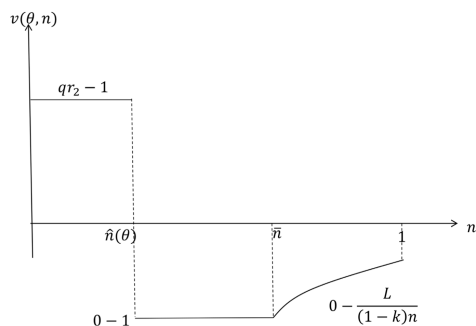


Fig. 3. Depositors' payoff differential when $(1-k) > L$. The figure shows a depositor's payoff differential between withdrawing at date 2 and at date 1 as a function of the proportion of depositors withdrawing early, n , when $1-k > L$. When $0 \leq n \leq \hat{n}(\theta)$, the bank has enough resources to make the promised repayments at both date 1 and 2. Thus, a depositor expects to receive r_2 with probability q at date 2, and 1 at date 1. When $\hat{n}(\theta) < n \leq 1$, the bank can still pay 1 at date 1, while fails to repay depositors at date 2. Finally, when $n < \hat{n} \leq 1$, the bank fails to make the promised repayment at both dates. Hence, a depositor receives nothing at date 2, while a share of the bank's liquidation proceeds at date 1.

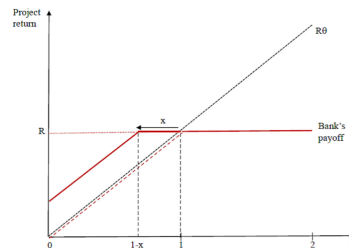


Fig. 4. Bank payoff with a first-loss guarantee as a function of the fundamental θ . The figure shows the bank's payoff as a function of θ in the presence of a first-loss guarantee of size x . This is equal to $R(\theta + x)$ for $\theta < 1-x$ and to R for $\theta \geq 1-x$.

5.3 Figure 3: Depositor payoff differential when $1 - k > L$

- In this case the payoff from waiting falls as more depositors withdraw.
- This creates the strategic complementarity that generates **panic runs**.
- Read the figure as the graphical counterpart of θ^* : low-capital banks are exposed to a wider run region because depositors fear one another's withdrawals.

5.4 Figure 4: Bank payoff with a first-loss guarantee

- The guarantee shifts the bank's payoff up in bad states from $R\theta$ to $R(\theta + x)$, until the bank reaches the full-repayment region.
- This figure makes the positive incentive effect intuitive: bad states become less bad *for the bank*, so surviving to date 2 becomes more valuable.
- That is the source of the paper's charter-value logic.

5.5 Figure 5: Evergreening without and with a guarantee

- The figure compares the bank's private continuation region with the social planner's liquidation region.
- With a guarantee, the bank is willing to continue projects over a larger set of low- θ states.
- So the figure illustrates the paper's central trade-off:

better ex ante monitoring versus more ex post evergreening.

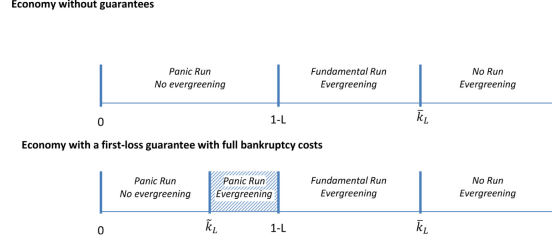


Fig. 5. Evergreening without and with a first-loss loan guarantee with full bankruptcy costs. Without guarantees, banks with $k < 1 - L$ are subject to panic runs and do not engage in evergreening, while banks with $k > 1 - L$ do so both when they are subject to fundamental runs for $K \in [1 - L, k_L]$, and when they are not subject to runs for $k \in (k_L, 1]$. With a first-loss guarantee with full bankruptcy costs, also banks with capital $k \in [k_L, 1 - L]$ start engaging in evergreening.

6 Short summary

- **Method.** A bank-run model with endogenous monitoring, bank capital, and multiple guarantee designs.
- **Mechanism.** Loan guarantees change both bank solvency and depositor run incentives, so underwriting and fragility are jointly determined.
- **Main result.** First-loss loan guarantees usually **improve monitoring** and reduce runs; deposit insurance does not.
- **Important qualification.** If guaranteed transfers are protected from bankruptcy, very poorly capitalized banks may monitor less.
- **Policy message.** Guarantee design matters: the same guarantee that stabilizes credit can also weaken market discipline or worsen evergreening.

7 Conclusion

7.1 Core conclusion

- The paper shows that the effect of loan guarantees on bank behavior cannot be evaluated using a simple moral-hazard argument.
- In a fragile banking structure with demandable deposits, loan guarantees often **increase** monitoring because they reduce run risk and raise the bank's continuation value.
- The major exception is the case of very poorly capitalized banks when guaranteed transfers remain available in bankruptcy and thereby mute depositor discipline too strongly.

7.2 Economic intuition

- A bank monitors more when the payoff from surviving to date 2 is larger.
- A loan guarantee can enlarge that survival value because it cushions default states.
- But the same guarantee can also make depositors less sensitive to bank risk, and that weakens discipline.

- The net effect therefore depends on **capital** and on **who gets the guarantee in insolvency**.

7.3 Takeaway

This paper's main lesson is that **loan guarantees and deposit insurance are not equivalent policy tools**. Loan guarantees can strengthen bank underwriting because they operate through the asset side of the balance sheet and interact with run risk. But that beneficial effect is not automatic: it depends on the bank's capitalization and on the exact legal design of the guarantee.